

Foundation (Jalapeno)

- Q1.** What is the equation of a line in point-slope form that passes through the point $(2, 3)$ with a slope of 4?
- (A) $y - 3 = 4(x - 2)$
(B) $y - 2 = 3(x - 4)$
(C) $y - 4 = 2(x - 3)$
(D) $y - 1 = 5(x - 2)$
(E) $y - 5 = 1(x - 1)$
- Q2.** A space shuttle is traveling at a rate of 250 miles per hour. If it travels for t hours, how many miles will it have traveled when $t = 5$ hours, given the equation $d = 250t$ is in slope-intercept form?
- (A) 1250 miles
(B) 1500 miles
(C) 1750 miles
(D) 2000 miles
(E) 2250 miles
- Q3.** Which equation represents a line in standard form with a slope of -2 and a y -intercept of 5?
- (A) $2x + y = 5$
(B) $2x - y = -5$
(C) $x + 2y = 5$
(D) $x - 2y = -5$
(E) $y = -2x + 5$
- Q4.** The equation $3x - 2y = 7$ is in which form?
- (A) Point-slope form
(B) Slope-intercept form
(C) Standard form
(D) None of the above
(E) All of the above
- Q5.** The graph of the equation $y = -3x + 2$ has a y -intercept of?
- (A) $(0, 2)$
(B) $(2, 0)$
(C) $(0, -2)$
(D) $(-2, 0)$
(E) $(0, 3)$
- Q6.** A physicist measures the temperature of a lab sample, finding it to be 25°C . The sample's temperature increases at a rate of 2°C per minute. Which equation models this situation, where t is time in minutes?
- (A) $T = 2t + 25$
(B) $T = 2t - 25$
(C) $T = -2t + 25$
(D) $T = 25t + 2$
(E) $T = 25t - 2$
- Q7.** A line passes through the points $(1, 4)$ and $(3, 5)$. What is its equation in slope-intercept form?
- (A) $y = \frac{1}{2}x + \frac{7}{2}$
(B) $y = \frac{1}{2}x - \frac{7}{2}$
(C) $y = \frac{3}{2}x + \frac{1}{2}$
(D) $y = \frac{3}{2}x - \frac{1}{2}$
(E) $y = \frac{1}{3}x + \frac{4}{3}$
- Q8.** Converting the equation $y = 2x - 5$ to point-slope form with the point $(1, -3)$ gives?
- (A) $y + 3 = 2(x - 1)$
(B) $y - 3 = 2(x + 1)$
(C) $y - 2 = 1(x - 3)$
(D) $y + 2 = -1(x - 3)$
(E) $y - 5 = 2(x - 1)$

Open Ended Questions (FRQ)

- Q9.** Graph the equation $y = -\frac{3}{2}x + 2$ and identify its x - and y -intercepts.
- Q10.** A new planet's atmospheric pressure P is related to its altitude h (in kilometers) above the planet's surface by the equation $P = 10 - 2h$. Convert this equation to standard form and then graph it, labeling the x - and y -intercepts. What is the atmospheric pressure at an altitude of 2 kilometers?

Chef's Tips

- Tip 1: Provide a graphing calculator
- Tip 2: Monitor student calculations for accuracy
- Tip 3: Encourage students to check their work